

and protein synthesis—also occur during quiet waking periods. Furthermore, in many ways sleeping is different from resting to catch your breath after extensive exercise.

- People need only a little more sleep after a day of extreme physical or mental activity than after a day of inactivity (Horne & Minard, 1985).
- Some people get by with much less than the “normal” 7½ to 8 hours of sleep per day. One extreme case was a healthy 70-year-old woman who slept only about 1 hour each night (Meddis, Pearson, & Langford, 1973).
- Some people fare better than we would have expected after a week or more of sleep deprivation (Figure 5.7). In 1965, a San Diego high-school student, Randy Gardner, stayed awake for 264 hours and 12 minutes—11 days—in a project for a high-school science fair. Gardner suffered no serious psychological consequences (Dement, 1972). On the last night of his ordeal, he played about 100 arcade games against sleep researcher William Dement and won every game. Just before the end of the 264 hours, he held a television press conference and handled himself well. After sleeping for 14 hours and 40 minutes, he awoke refreshed.

If a torturer prevented you from sleeping for the next 11 days, would you do as well as Randy Gardner? Probably not, for two reasons: First, Gardner knew he was in control of the situation and that he could quit if necessary. A torturer would not give you that option. Second, people vary in their ability to tolerate sleep deprivation. We heard about Gardner only because he tolerated it so well. We have no idea how many other people tried to deprive themselves of sleep for several days but gave up.

The summary evaluation of the repair and restoration theory is that the body does do some restoration during sleep, and sleep deprivation is harmful, although more so to some people than to others. However, each of us has an urge to sleep, even a need to sleep, at a certain time of day, regardless of how active or inactive we have been during the day. Also, most of the restorative functions of sleep could occur during occasional quiet times, without setting aside an 8-hour period for sleep. Therefore, it is possible that sleep serves other functions as well.

The Evolutionary or Energy-Conservation Theory

Sleep is a way of conserving energy. If we built a solar-powered robot to explore another planet, we would probably program it to shut down most of its activities at night. According to the **evolutionary, or energy-conservation, theory of sleep**, *evolution equipped us with a regular pattern of sleeping and waking for the same reason—to conserve fuel and to prevent us from walking into dangers* (Kleitman, 1963; Webb, 1979). Every species has an inefficient time of day; night is that time for humans and other highly visual species. Throughout our evolutionary



FIGURE 5.7 Even near the end of Randy Gardner’s 264 consecutive hours without sleep, he was able to perform tasks requiring strength and skill. Here, observers dutifully record his every move.

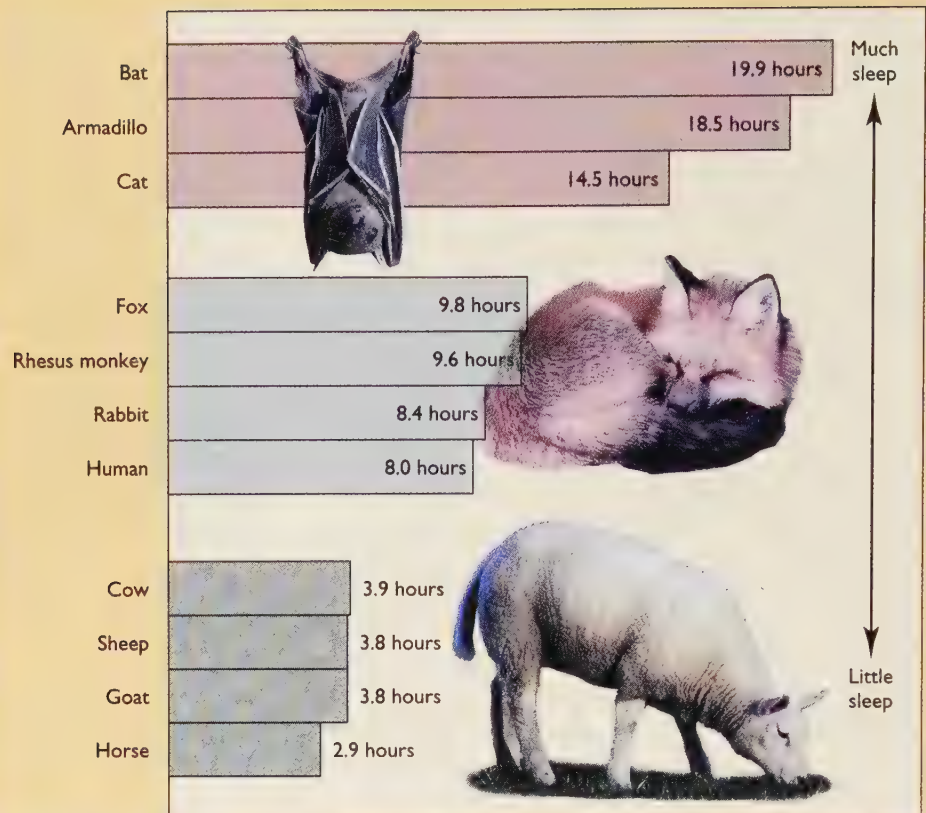
history, until the invention of electric lights, any activity conducted at night was likely to be wasteful and possibly dangerous. By sleeping, we decrease our energy use by 10 to 25%. At times when food is especially scarce, we either sleep more or we lower our body temperature more than usual, thus conserving even more energy (Berger & Phillips, 1995).

However, sleep protects us only from the sort of trouble we might walk into; it does not protect us from trouble that comes looking for us! So, we sleep well when we are in a familiar, safe place; but we sleep lightly, if at all, when we fear that burglars will break into the house or that bears will nose into our tent.

The evolutionary theory accounts well for differences in sleep among species (Campbell & Tobler, 1984). For example, why do cats sleep so much and horses so little? First, cats can afford to have long periods of inactivity because they eat one or two large, energy-rich meals per day; horses need to spend many hours grazing every day. Second, cats are unlikely to be attacked while they sleep, whereas horses must always be ready to run away from a predator (Figure 5.8). (Woody Allen once said, “The lion and the calf shall lie down together, but the calf won’t get much sleep.”)

Which of these two theories regarding sleep is correct? Both are, to a large degree. Supporters of the repair and restoration theory concede that the timing and even the amount of sleep depend on when the animal is least efficient at finding food and defending itself. Supporters of the evolutionary theory concede that, during the time

FIGURE 5.8 Sleep time for mammals varies widely. Animals that are rarely attacked sleep a lot; those in danger of attack sleep only a few hours. Diet also relates to sleep. (Based on data from Zepelin & Rechtschaffen, 1974.)



that evolution has set aside for an animal to conserve energy, that animal takes that opportunity to perform repair and restoration functions.

Stages of Sleep

In the mid-1950s, the French scientist Michel Jouvet (Jouvet, Michel, & Courjon, 1959) and the American researchers William Dement and Nathaniel Kleitman (1957a, 1957b) independently discovered a stage of sleep called either *paradoxical sleep* or **rapid-eye movement (REM) sleep**. *During this stage of sleep, the sleeper's eyes move rapidly back and forth under the closed lids.* (All other stages of sleep are known as non-REM, or NREM, sleep.) A paradox is an apparent contradiction; REM sleep is sometimes called paradoxical because it is light in some ways and deep in others. It is light because the brain is active and the body's heart rate, breathing rate, and temperature fluctuate substantially (Parmeggiani, 1982). It is deep because the large muscles of the body, those that control posture and locomotion, are very relaxed. It also has some features that are hard to classify as either deep or light; for example, males typically have an erection of the penis and females have lubrication of the vagina during REM sleep.

People are more likely to report dreams, especially vivid and highly visual dreams, if they are awakened during REM sleep than if awakened during NREM sleep, although some sort of dream report is common in both states. The discovery of REM sleep inspired much research interest in dreaming.

Sleep Cycles During the Night

Sleep researchers distinguish among stages of sleep by recording brain waves with electrodes attached to the scalp (Figure 5.9). A device called an **electroencephalograph**, abbreviated **EEG**, *measures and amplifies slight electrical changes on the scalp that reflect patterns of activity in the brain*. Sleep researchers combine an EEG measure with a simultaneous measure of eye movements to produce a **polysomnograph** (literally, "many-sleep measure") as shown in Figure 5.10. Upon falling asleep, one enters sleep non-REM stage 1, when the eyes are nearly motionless and the EEG shows many short, choppy waves, as shown in Figure 5.10a. These small waves indicate a fair amount of brain activity, with brain cells firing out of synchrony with one another. Because they are out of synchrony, their activities nearly cancel each other out, like a crowd of people talking at the same time.

As sleep continues, a person progresses into non-REM stages 2, 3, and 4, as shown in Figure 5.10b–e. These stages

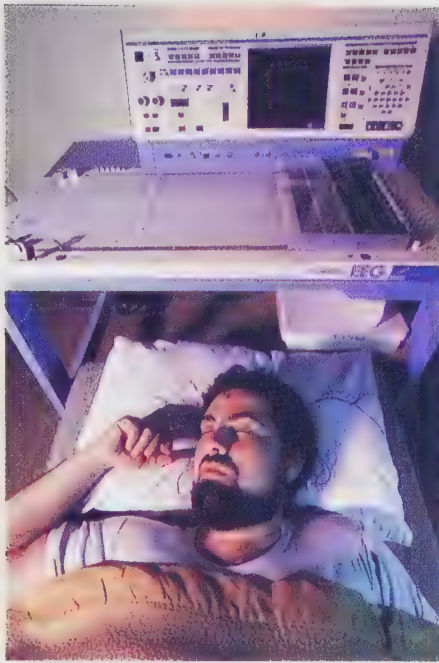


FIGURE 5.9 Electrodes monitor the activity in a sleeper's brain, and an EEG then records and displays brain-wave patterns.

are most easily distinguished by the number of long, slow waves; stage 1 has the fewest, and stage 4 has the most. These large waves indicate *decreased* brain activity. They grow larger from one stage to the next, because a larger proportion of the active neurons are active at the same time. (Note that the eyes remain mostly inactive in each of these stages.)

After stage 4, a sleeper gradually moves back through stages 3 and 2. After that, however, the sleeper enters REM sleep, not stage 1. That is, REM sleep replaces most of stage 1 except for the first episode of the night. In Figure 5.10f, the EEG in REM sleep resembles that of stage 1, but the eyes are moving more extensively and more regularly than they do during any other stage. At the end of REM sleep, the sleeper cycles through stages 2, 3, 4, and then back to 3, 2, and REM again. In a healthy young adult, each cycle lasts 90 to 100 minutes. As shown in Figure 5.11, the cycles continue throughout the night, but as the night progresses, the duration of stages 3 and 4 gets shorter and the duration of REM and stage 2 increases. Figure 5.11 represents sleep under quiet, undisturbed conditions; if the room is noisy, too hot, too cold, or uncomfortable in any other way, the person will have many awakenings and whatever sleep the person does manage will be mostly stage 2 (Veluti, 1997).

Sleep Stages and Dreaming

William Dement's early research indicated that people who were awakened during REM sleep usually reported they had been dreaming but that people who were awakened

during any other period seldom reported dreaming. So, for a while, REM sleep was thought to be almost synonymous with dreaming. However, later studies found that much dreaming also occurs during non-REM sleep, although non-REM dreams are less vivid, less visual, less bizarre, and less likely to be experienced as something that's really happening.

The link between REM sleep and the more vivid dreams prompted sleep investigators to awaken people during REM sleep to ask for dream reports. Scientific progress frequently depends on an improved way of measuring something; in this case, a method for measuring dreaming enabled researchers to answer some basic questions.

For example, does everyone dream? People who claim they do not dream have been taken into the laboratory and attached to a polysomnograph, which revealed normal periods of REM sleep. When these people were awakened during a REM period, they reported dreams (to their own surprise). Apparently, they dream as much as anyone else but forget their dreams faster.

Another question: How long do dreams last? William Dement and Edward Wolpert (1958) awakened people after REM periods of various durations and asked them to describe their dreams. A person awakened after 1 minute of REM sleep would usually tell a brief story; a person awakened after 5 minutes of REM sleep would usually tell a story about 5 times as long, and so on. Evidently, dreams take place in "real time." That is, a dream is not over in a split second; if a dream seemed to last several minutes, it probably did.

Many other questions about dreaming remain unanswered, however. For example, how accurately do we remember our dreams? Memory distortions occur even for waking events; they are probably much more extreme for dreams, but we do not know how to measure this distortion. Or consider this apparently simple question: Do we dream in color? The fact that people ask this question reveals why it is difficult to answer: They ask because they do not remember whether their dreams were in color. But how could an investigator determine the answer except by asking people? The best evidence we have is that, when people are awakened during REM sleep, when their recall should be as sharp as possible, they report color at least half of the time (Herman, Roffwarg, & Tauber, 1968; Padgham, 1975). This result does not necessarily mean that their other dreams are in black and white; it may mean only that the colors in those dreams were not memorable.

CONCEPT CHECK

3. Is dreaming more common toward the end of the night's sleep or toward the beginning? (Check your answer on page 172.)

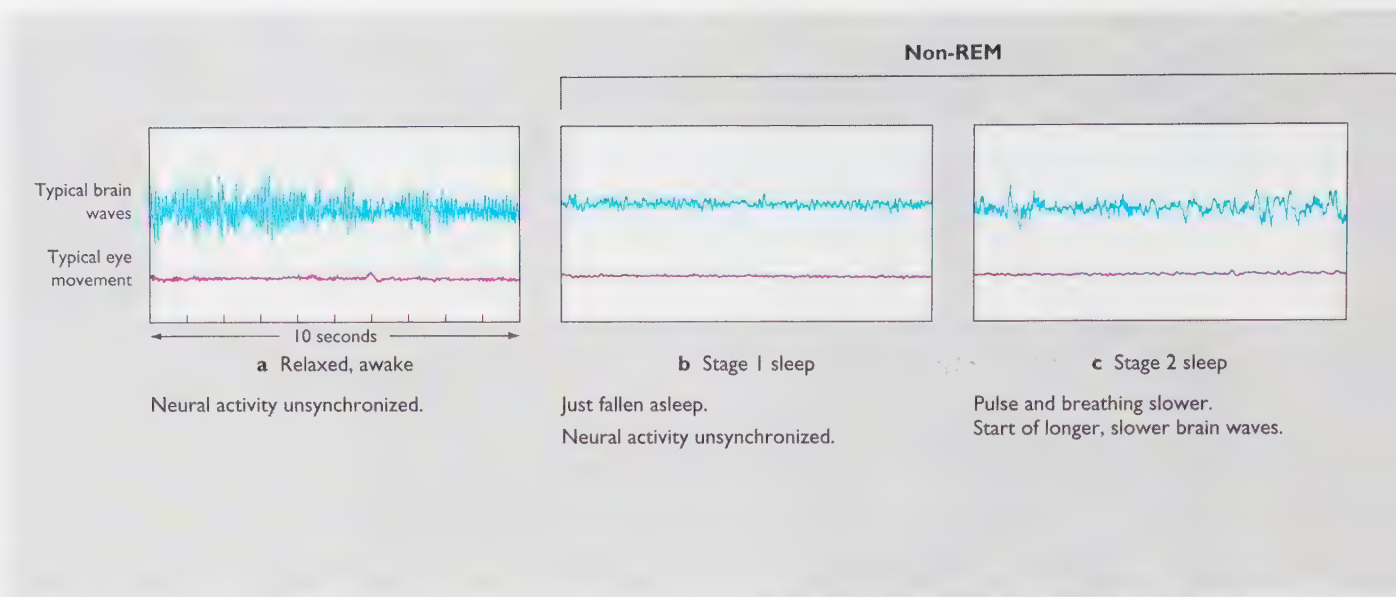


FIGURE 5.10 During sleep, people progress through stages of varying brain activity. The blue line indicates brain waves, as shown by an EEG. The red line shows eye movements. Note that REM sleep resembles stage 1 sleep except for the addition of rapid eye movements. (Courtesy of T. E. Le Vere.)

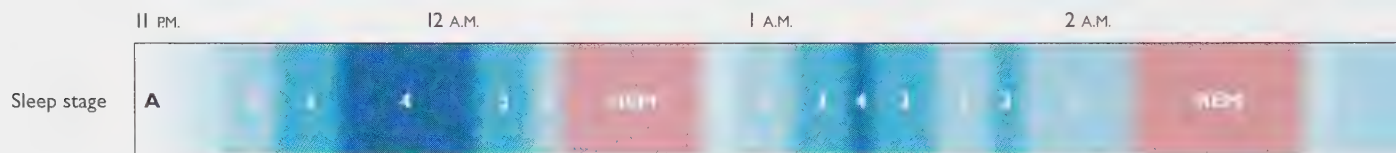


FIGURE 5.11 This fairly typical sleeper had five cycles of REM and non-REM sleep and awakened (A) briefly three times during the night. Note that stage 4 occupies much time early in the night but less later; REM sleep becomes more and more prevalent as the night progresses. (From Dement, 1972.)

The Functions of REM Sleep

Given that people spend 20–25% of an average night in REM sleep, it presumably serves an important function, but what? The most direct way to approach this question is to deprive people of REM sleep to study how this deprivation affects their health or behavior.

In one study, William Dement (1960) monitored the sleep of eight young men for seven consecutive nights and awakened them for a few minutes whenever their EEG and eye movements indicated the onset of REM sleep. He awakened the members of a control group equally often but at random times, so that he did not necessarily interrupt their REM sleep. Over the course of one week, Dement found it harder and harder to prevent REM sleep in the experimental group. On the first night, the average subject had to be awakened 12 times; by the seventh night, 26 times. During the day, REM-deprived subjects experienced anxiety, irritability, and impaired concentration. On the eighth night, all the subjects were permitted to sleep without interruption. Most of them showed a “REM rebound,” spending 29% of the night in REM sleep as compared with 19% before the experiment began. The subjects in the

control group, who had not been deprived of REM sleep, showed no such REM rebound.

So, evidently, REM sleep does satisfy a need, and the body will work to catch up on at least some of the REM sleep that it has been forced to miss. However, in this and related studies, the effects of REM deprivation have not been catastrophic, so the question of why we need REM sleep remained.

A second approach to the question is to determine which people get more REM sleep than others. One clear pattern is that infants get more REM sleep than children, and children more than adults. From that observation, many researchers have inferred that REM sleep serves a function that is more acute in younger people. Maybe so, but we should be cautious when interpreting this evidence. Infants not only get more REM sleep but also get more total sleep (Figure 5.12). If we compare species, we find that the species that get the most sleep (such as cats) also generally have the greatest percentage of REM sleep. Among adult humans, those who sleep 9 or more hours per night, spend a large percentage of that time in REM sleep; those who sleep 6 hours or less spend a smaller percentage in REM sleep. In short, the individuals with the greatest



amount of total sleep time spend the greatest percentage of that time in REM. It is as though a certain amount of *non-REM* sleep is necessary each night; additional amounts of REM sleep can be added if the period of sleep continues long enough (Horne, 1988).

A growing body of research suggests that one function of REM sleep, though not necessarily the main one, is to improve memory storage. When animals are given new learning experiences, they generally get more than the usual amount of REM sleep the following night, and ani-

mals prevented from getting REM sleep learn slowly (Hobson, 1992). As a rule, the animals that learn fastest are also the ones that show the greatest increase in REM sleep (Smith & Wong, 1991). In humans, research indicates that REM sleep is more important for storing some kinds of memories than others. Young adults in one experiment memorized some words and learned a motor skill—tracing some pictures as seen in a mirror—just before a period of sleep either early in the night (mostly non-REM sleep) or late in the night (mostly REM sleep). Other students

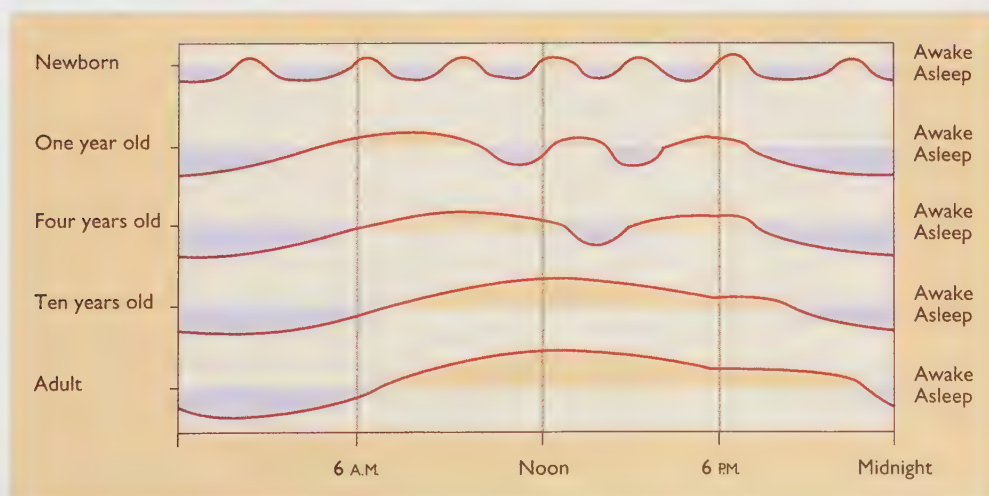


FIGURE 5.12 Newborns' sleep alternates between wakefulness and naps throughout the day. Within a few months, infants consolidate most of their sleep into one longer period at night, although they continue having one or two naps during the day. As people grow older, the amount of sleep per day decreases. (Based on Kleitman, 1963.)

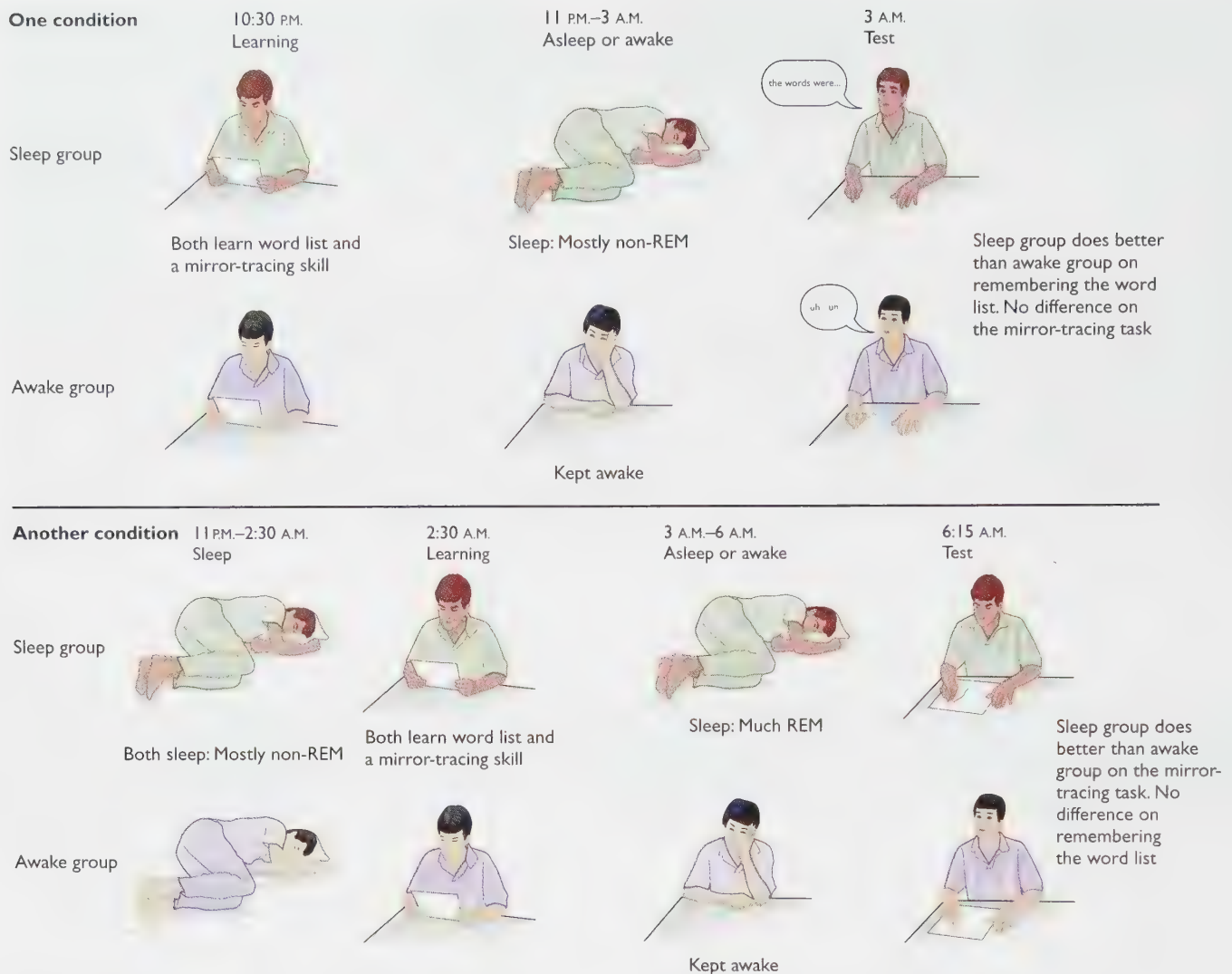


FIGURE 5.13 Students who learned materials before a period of mostly non-REM sleep thus improved their memory of verbal materials but not motor skills. Those who learned just before a period of mostly REM sleep improved their memory of motor skills but not verbal materials. (Based on results of Plihal & Born, 1997.)

learned the same materials but were then required to stay awake. After three hours of sleep or wakefulness, all were tested for their memory of the two tasks. Those who had experienced three hours of mostly non-REM sleep showed enhanced memory of the word lists. Those who had mostly REM sleep showed enhanced memory of the motor skills (Plihal & Born, 1997). Figure 5.13 summarizes this experiment. The apparent conclusion is that REM sleep is specifically important for strengthening the memories of motor skills but not memories of factual or verbal materials.

The Content of Our Dreams

Even a saint is not responsible for what happens in his dreams.
—ST. THOMAS AQUINAS

What do we dream about and why? People once believed that dreams foretold the future. Occasionally, of course, they do—either by coincidence or because the dreamer

had a reason to expect a certain outcome. Sigmund Freud maintained that dreams reveal a person's unconscious thoughts and motivations. Although Freud's interpretations often revealed more about Freud's imagination than about the dreamer, people's dreams usually do relate somehow to recent thoughts and concerns, though not necessarily unconscious ones (Arkin & Antrobus, 1978). For example, people deprived of fluids frequently dream about drinking; people who have been kept in isolation dream about talking in groups. The dreams of old people differ from those of the young; dreams of women differ from those of men. Curiously, women dream about men more than men dream about women (Domhoff, 1996; Hall & Van de Castle, 1966).

For adult dreamers in the United States—and apparently in other countries, too, according to more limited data—one common dream theme is “things that could go wrong.” People often dream of falling, being chased, being naked or almost naked in public, or being unable to do something they need to do. The most common emotion in

dreams is apprehension or fear; more dreams include misfortune than good fortune; and the dreamer is more often the victim of aggression than the cause of it (Domhoff, 1996; Hall & Van de Castle, 1966). People occasionally awaken with disappointment that it was only a dream, but more often they awaken with relief. A frequently recurring dream is often a sign of a persistent worry or concern. People who have the same dream over and over tend to report more anxiety, depression, and stress in their waking lives than do other people (Brown & Donderi, 1986).

Dreams are a product of brain activity, and even a sleeping brain experiences a fair amount of sensory information. While you sleep, you can hear any noises in the room, and your vestibular system detects the position of your head (Velluti, 1997). Sometimes these stimuli affect dreams; for example, a noise that is not loud enough to awaken someone might become an explosion or an alarm in a dream. Usually, however, the content of a dream is only remotely related to stimuli in the room, if at all.

A more important correlate of dreams is the spontaneous activity that arises in the brain during REM sleep, particularly in the visual areas. According to the **activation-synthesis theory of dreams**, *the brain experiences this spontaneous activity as sensations, links the sensations together, and tries to synthesize them into a coherent pattern* (Hobson, 1988). A dream is your brain's best effort to make sense of the limited information it is receiving.

For example, the visual areas of the brain become active during REM sleep, especially during the first minutes of a REM period, so nearly all dreams include visual content (Amzica & Steriade, 1996). In one study, a man repeatedly slept in a laboratory; the investigators awakened him during REM periods. They found that the more eye movements he had during his REM periods, the more visual imagery he reported in his dreams, especially imagery of watching other people move (Hong et al., 1997). However, this was just a single study with one participant. Recall from Chapter 2 that we cannot draw cause-and-effect conclusions from correlational studies; that is, we cannot say whether the eye movements controlled the dreams, the dreams affected the eye movements, or another factor controlled both of them.

You might wonder whether blind people have visual dreams. The answer depends on when and how a person became blind. People who had visual experience before eye damage continue to see during their dreams. (The visual cortex is still intact and becomes spontaneously active during REM sleep.) But a person who never had vision or who lost it because of damage to the visual cortex has no visual imagery during dreams, only auditory and touch imagery. People with any degree of visual impairment are more likely than sighted people to dream about touching something (Sabo & Kirtley, 1982).

Many other kinds of dreams also appear to relate to the spontaneous activity of the brain during REM sleep. Many people occasionally dream of flying or falling, probably because the vestibular system detects the body's horizontal

position and the brain interprets this sensation as floating or flying (Hobson & McCarley, 1977). Many dreams include strong emotions, especially fear; the *amygdala*, a brain area known to be linked to fear, is highly active during REM sleep (Maquet et al., 1996). Have you ever dreamed that you are trying to walk or run away but you cannot move? One possible reason for this is that the major postural muscles are really paralyzed during REM sleep. Thus, your brain could send messages telling your muscles to move but then receive sensory feedback indicating that they have not moved at all. Finally, consider that many dreams jump incoherently among unrelated scenes and events. One possible explanation is that the prefrontal cortex, which is important for storing recent memories, is substantially *less* active than usual during REM sleep (Maquet et al., 1996).

The problem with the activation-synthesis theory, however, is that it does not make clear, testable predictions. For example, people almost always sleep horizontally but only occasionally dream of flying or falling: Why not always? Our muscles are always paralyzed during REM sleep: Why do we sometimes dream that we're moving but other times dream that we can't? *After* a dream, the activation-synthesis theory can provide explanations, but it offers only vague predictions of who will dream about what and when. This theory can therefore be criticized for not being falsifiable (see Chapter 2).

Finally, however, the activation-synthesis theory does not necessarily imply that our dreams are meaningless. Even if they begin with more-or-less random activity in various brain areas, the dreamer's interpretations of this activity depend on his or her personality, motivations, and previous experiences.

Abnormalities of Sleep

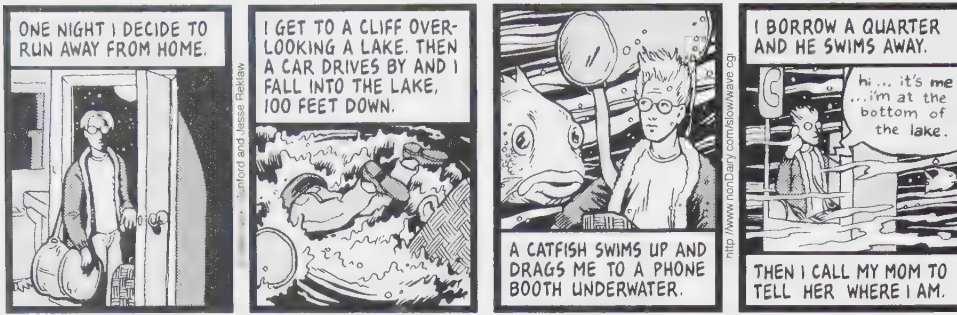
Comedian Steven Wright says that someone asked him, "Did you sleep well last night?" He replied, "No, I made a few mistakes."

We laugh because sleep isn't the kind of activity on which a person makes mistakes; sleep just happens. Sometimes, however, sleep doesn't happen, or it happens at the wrong time, or it does not seem restful, or we have bad dreams. We would not call these unpleasant experiences "mistakes," but in one way or another, our sleep is not what we wanted it to be.

Insomnia

The term *insomnia* literally means "lack of sleep." However, we cannot usefully define insomnia in terms of the number of hours of sleep. Some people feel well rested after less than 6 hours of sleep per night; others feel poorly rested after 9 hours. A complaint of **insomnia** indicates that *the person feels poorly rested the next day*. By this definition,

SLOW WAVE



SLOW WAVE



These comic strips represent actual dreams as described to the artist. Dreams often mix possible with impossible events and frequently explore the theme of “things that could go wrong.”

about one third of all adults have occasional insomnia and about one tenth have serious or chronic insomnia (Lilie & Rosenberg, 1990). Most of those with serious insomnia have other medical or psychological disorders as well, such as anxiety disorders or depression (Ohayon, 1997).

People sometimes have trouble sleeping because of noise, worries, indigestion, uncomfortable temperatures, use of alcohol or caffeine, and other miscellaneous problems. Overuse of tranquilizers can also cause insomnia. That statement may be surprising, because people often take tranquilizers as a way of *relieving* insomnia. The problem is that no pill will exert its effects for exactly the period of time that someone wants to sleep. Some tranquilizers produce brief effects that wear off before morning, so the person awakens early (Kales, Soldatos, Bixler, & Kales, 1983). Others have effects that last too long, so the person remains sleepy for part of the next day. Moreover, people who use tranquilizers as sleeping pills may come to depend on them, so that they cannot get to sleep without them (Kales, Scharf, & Kales, 1978).

Disorders Related to Insomnia

Most people with insomnia suffer from a variety of other complaints and disorders, including tension, depression, personality disturbances, loss of energy, and increases in

body temperature and metabolism. Does the lack of sleep *cause* these other problems? Maybe, maybe not; the fact that people with insomnia also tend to have other problems is merely a correlation, and a correlation does not demonstrate causation.



Insomnia is identified not by how many hours one sleeps at night but by how sleepy the person is the following day.

How might we determine whether insomnia leads to these other problems? Here is a clever experiment that addresses this issue (Bonnet & Arand, 1996).

Hypothesis A previously normal person who is forced to sleep only as long as someone with insomnia, at the same times, will develop a similar pattern of medical and psychological abnormalities.

Method Experimenters identified 10 adults with insomnia, who were defined as people who, for at least the last year, on more nights than not, took at least 45 minutes to get to sleep and awoke for at least an hour during the night. The experimenters then advertised for volunteers who had no sleeping problems. For each person with insomnia, they matched one volunteer of the same gender and approximately the same age and weight. For seven nights, the experimenters monitored the sleep of the people with insomnia. Whenever one of them went to sleep, the volunteer partner was allowed to go to sleep. When the person with insomnia woke up, the matched volunteer was awakened also. Thus, each volunteer awoke and went to sleep at the same times as one person with insomnia. This kind of design is known as a *yoked control experiment*, because each participant in one group is “yoked” to one person in the other group. At various times during the week, each person was given a variety of medical and psychological tests.

Results Table 5.1 summarizes the results.

Interpretation After one week of enforced insomnia, the normal people had less vigor, so we may conclude that lack of sleep decreases pep and energy. In other regards, however, the week of enforced insomnia produced either no effect or effects opposite to those found in people suffering from chronic insomnia. We conclude, therefore, that sleep deprivation by itself does not cause tension, depression, personality disturbances, or increases in body temperature or metabolic rate. Instead, many people develop insomnia because of nervous system disorders that also lead to tension, depression, high body temperature, and so forth.



Sleep Apnea

Among the many possible causes of insomnia, one is known as *sleep apnea* (AP-nee-uh). *Apnea* means “no breathing.” Many people have either irregular breathing or occasional periods of 10 seconds or so without breathing during their sleep. People with **sleep apnea**, however, may *fail to breathe for a minute or longer and then wake up gasping for breath*. When these people do manage to breathe during their sleep, they generally snore. They may lie in bed for 8 to 10 hours per night but actually sleep less than half that time. During the following day, they are likely to feel sleepy and they may have headaches.

Many people with sleep apnea are obese—especially obese middle-aged or older men, who are unable to find a sleeping position that lets them breathe easily (Mezzanotte, Tangel, & White, 1992). Others have brain abnormalities, especially in the medulla, that interfere with their breathing during sleep.

Narcolepsy People who experience *sudden attacks of extreme, even irresistible, sleepiness in the middle of the day* are said to have **narcolepsy** (Aldrich, 1993). Such people may also experience sudden attacks of muscle weakness or paralysis. Sometimes they have vivid dreamlike experiences while they are awake. Each of these symptoms could be interpreted as a sudden intrusion of sleep, especially REM sleep, into the waking period of the day.

Narcolepsy is a rare condition, affecting about one person per thousand. Its causes are unknown, although it often runs in families. Physicians prescribe stimulants and antidepressant drugs, both of which are helpful in many cases. Some physicians are hesitant to prescribe these drugs, however, because of side effects and the potential for abuse of stimulant drugs (Kryger, 1993).

Sleep Talking, Sleepwalking, Nightmares, and Night Terrors

Many people, whether or not they suffer from insomnia, nevertheless have unsettling experiences during their sleep. Sleep talking is the most common and least troublesome.

TABLE 5.1 Test Results in Insomnia Experiment

	PEOPLE WITH INSOMNIA	NORMAL PEOPLE AFTER 1 WEEK OF ENFORCED INSOMNIA
<i>Tension</i>	More tense than most people	Decreased tension
<i>Mood</i>	Depressed	Not depressed
<i>Vigor</i>	Low	Low
<i>Personality</i>	Disturbed in several ways	Normal
<i>Estimate of own sleep</i>	Underestimate amount of sleep	Correctly estimate amount of sleep
<i>Body temperature</i>	Higher than normal	Lower than normal
<i>Metabolic rate</i>	Higher than normal	Lower than normal

Most people talk in their sleep more often than they realize, because they do not remember sleep talking themselves and usually no one else is awake to hear them. Sleep talking occurs with about equal probability in REM sleep and non-REM sleep. It can range from a single, indistinct word or grunt to a clearly articulated paragraph. Sleep talkers sometimes pause between utterances, as if they were carrying on a conversation with someone else. In fact, it is possible to engage some sleep talkers in a dialogue. Sleep talking is nothing to worry about. It is not related to any mental or emotional disorder, and sleep talkers rarely say anything that they would be embarrassed to say when awake.

Sleepwalking tends to run (walk?) in families. True sleepwalking occurs mostly in children during stage 4 sleep and lasts less than 15 minutes. Few children hurt themselves when sleepwalking, and most children outgrow it (Dement, 1972). A person who appears to be sleepwalking may really be awake but confused. You have no doubt heard people say that you should never awaken a sleepwalker. This is another of those misconceptions, like “we use only 10 percent of our brain,” in which people are quoting each other, each person confident that the others know what they are talking about. In fact, sleep researchers report that waking a sleepwalker is neither dangerous nor harmful, although the person may indeed be disoriented and confused (Moorcroft, 1993).

Other concerns include nightmares and night terrors. A nightmare is an unpleasant dream, but a dream nevertheless. A night terror, however, is a state of extreme panic, including a heart rate three times the normal rate. Night terrors occur during stage 3 or stage 4 sleep but never during REM sleep. They are fairly common in young children and less common in adults (Salzarulo & Chevalier, 1983).

Leg Movements While Trying to Sleep

Have you ever lain in bed, trying to fall asleep, when suddenly one of your legs kicked? An occasional leg jerk while trying to fall asleep is a common experience and no cause for concern. In contrast, some people have *prolonged “creepy-crawly” sensations in their legs, accompanied by repetitive leg movements strong enough to awaken the person, especially during the first half of the night* (Moorcroft, 1993). This condition, known as **periodic limb movement disorder** (or, more informally, as restless leg syndrome), is a common cause of poor sleep in people over the age of 50. Sufferers can stop the leg kicks by standing and walking, but by doing so they of course interrupt their sleep. The causes of this disorder are unknown, and the best advice is to avoid factors that can make the condition worse—such as caffeine, stress, or fatigue. Tranquilizers suppress these leg movements in some people (Schenck & Mahowald, 1996).

CONCEPT CHECK

4. Why would it be unlikely, if not impossible, for sleepwalking to occur during REM sleep? (Check your answer on page 172.)

Hypersomnia

Finally, consider a disorder that is the opposite of insomnia: A small number of people suffer from **hypersomnia, excessive but unrefreshing sleep**. In a typical case, the person sleeps 8 or 9 hours per day during the week, plus an afternoon nap, and then “catches up” by sleeping 18, 20, even 22 hours per day on weekends. Even after this much sleep, the person has trouble awakening and feels confused and poorly rested. In some cases, the disorder can be traced to a history of head trauma, viral infection, or mood disorder, but in many cases the cause is unknown (Bassetti & Aldrich, 1997).

If You Have Trouble Sleeping . . .

Insomnia can be a brief, minor annoyance or a sign of a potentially serious disorder. If you experience persistent difficulties, consult a physician, but for occasional or minor insomnia you can try a few things yourself (Hauri, 1982; Lillie & Rosenberg, 1990):

- Try to associate your bed with sleeping, not with lying awake. If you find that you just can’t get to sleep, get up, do something else, and return to bed when you start to feel sleepy.
- Try to wake up at the same time each day. (It is hard to get to sleep at a regular time if you don’t also wake up at a regular time.)
- Avoid caffeine and nicotine, especially in the evenings.
- Avoid habitual use of alcohol or sleeping pills. (Either may help you get to sleep occasionally, but they become counterproductive after repeated use.)
- Keep your bedroom cool and quiet.
- Get a consistent amount of exercise daily.



Sleep and dreams are unlike wakefulness in some ways, but they do not represent a state of complete unconsciousness. Sounds and other stimuli present in the room can become parts of a dream, and if the stimuli are sufficiently meaningful, such as the sound of a child crying, the dreamer will awaken. The brain is never completely off duty, never completely relaxed.

Although our understanding of sleep and dreams con-

tinues to grow, several major questions remain. Even such basic issues as the function of REM sleep remain in doubt. People have long found their dreams a source of wonder, and researchers continue to find much of interest and mystery.

S U M M A R Y

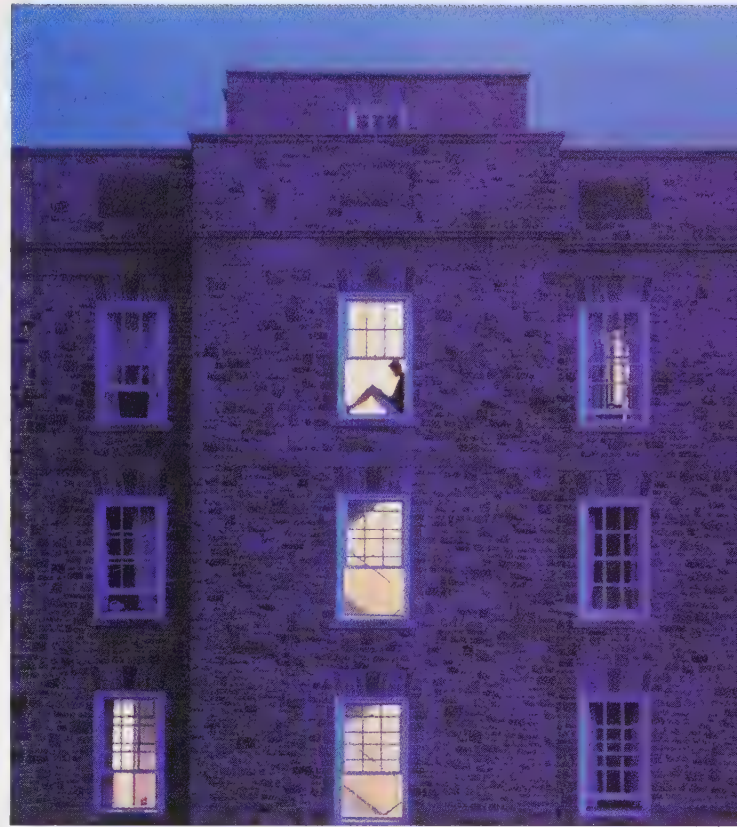
- ★ **Circadian rhythms.** Sleepiness depends on the time of day. Even in an unchanging environment, people become sleepy in cycles of approximately 24 hours. (page 157)
- ★ **Theories about our need for sleep.** Several repair and restoration functions take place during sleep. Sleep also serves to conserve energy at times of relative inefficiency. (page 160)
- ★ **Sleep stages.** During sleep, people cycle through sleep stages 1 through 4 and back through stages 3 and 2 to 1 again. The cycle beginning and ending with stage 1 lasts about 90 to 100 minutes. (page 162)
- ★ **REM sleep and dreams.** A special stage known as REM sleep replaces many of the stage-1 periods. REM sleep is characterized by rapid eye movements, a high level of brain activity, and relaxed muscles. People usually dream during this stage but can dream in other stages also. (page 163)
- ★ **Insomnia.** Insomnia—subjectively unsatisfactory sleep—can result from many influences, including a biological rhythm that is out of phase with the outside world, sleep apnea, narcolepsy, overuse of sleeping pills, and periodic limb movement disorder. (page 168)

Suggestions for Further Reading

- Dement, W. C. (1992). *The sleepwatchers*. Stanford, CA: Stanford Alumni Association. An account by one of the founders of sleep research.
- Moorcroft, W. (1993). *Sleep, dreaming, and sleep disorders: An introduction*. (2nd ed.). Lanham, MD: University Press of America. An excellent review of research on many aspects of sleep and dreams.

Terms

- circadian rhythm** a rhythm of activity and inactivity lasting approximately one day (page 157)
- jet lag** the discomfort and inefficiency that travelers experience in a new time zone because their internal clocks are out of phase with the light-dark cycle of their new environment (page 159)
- repair and restoration theory** the theory that the purpose of sleep is to enable the body to recover from the exertions of the day (page 160)
- evolutionary theory (or energy-conservation theory) of sleep** the theory that sleep evolved primarily as a means of forcing animals to conserve their energy when they are relatively inefficient (page 161)



One recommended strategy if you have trouble sleeping: Don't just lie in bed worrying about your lack of sleep. Get up, do something else, and try going back to sleep later.

- rapid eye movement (REM) sleep** a stage of sleep characterized by rapid eye movements, a high level of brain activity, and deep relaxation of the postural muscles; also known as paradoxical sleep (page 162)
- electroencephalograph (EEG)** a device that measures and amplifies slight electrical changes on the scalp that reflect brain activity (page 162)
- polysomnograph** a device that measures sleep stages using a combination of EEG and eye-movement records (page 162)
- activation-synthesis theory of dreams** the theory that parts of the brain are spontaneously activated during REM sleep and that a dream is the brain's attempt to synthesize these sensations into a coherent pattern (page 167)
- insomnia** failure to get enough sleep at night in order to feel well rested the next day (page 168)
- sleep apnea** a condition causing a person to have trouble breathing while asleep (page 169)
- narcolepsy** a condition characterized by suddenly falling asleep, or at least feeling very sleepy, during the day (page 169)
- periodic limb movement disorder** a condition occurring during sleep, marked by unpleasant sensations in the legs and many repetitive leg movements strong enough to interrupt sleep (page 170)
- hypersomnia** excessive but unrefreshing sleep (page 170)

Answers to Concept Checks

1. You should prefer to schedule the meeting on a Pacific island so that you will travel west and the other person will travel east. (page 159)
2. Try a melatonin pill two or three hours before your desired bedtime—in this case, about 1:00 to 2:00 P.M. (page 159)
3. REM sleep and dreaming are more common toward the end of the night's sleep. (page 163)
4. During REM sleep, the major postural muscles of the body are completely relaxed. (page 170)

Web Resources

SleepNet

shell5.ba.best.com/~ssm/

Learn more about sleep disorders and treatment, with links to many research reports and dream research and analysis sites; forums for discussing sleep and dreams; and The SnoozePaper, a monthly publication devoted to sleep in the news.

Hypnosis

What can hypnosis do?

What are its limitations?

*Truth is nothing but a path traced between errors.**

—FRANZ ANTON MESMER

If a hypnotist told you that you were 4 years old and you suddenly started acting like a 4-year-old, we would say that you are a good hypnotic subject. If the hypnotist said that you see your cousin sitting in the empty chair in front of you, and you said, “yes, I see her”; then again we would remark that you are deeply hypnotized.

But what if you had *not* been hypnotized and you suddenly started acting like a 4-year-old? or insisted that you saw someone in that empty chair? In this case, psychologists would suspect that you were suffering from a serious psychological disorder. Hypnosis induces a temporary state that is sometimes bizarre; no wonder we find it so fascinating.

Psychologists define **hypnosis** as *a condition of increased suggestibility that occurs in the context of a special hypnotist-subject relationship*. The term *hypnosis* comes from Hypnos, the Greek god of sleep. Although it has long been assumed that hypnosis is somehow related to sleep, the connection is only superficial: In both states the eyes are usually closed and the person is without initiative, and a hypnotized person, like a dreamer, accepts contradictory information without protest. A hypnotized person, however, can walk around and respond to objects and events in the real world. Also, the EEG of a hypnotized person is like that of a waking person with closed eyes, not like that of a sleeping person.

Hypnosis was first practiced by an Austrian philosopher and physician, Franz Anton Mesmer (1734–1815). When treating certain medical problems, Mesmer would pass a magnet back and forth across the patient’s body to redirect the flow of blood, nerve activity, and certain undefined “fluids.” Some of his patients reported dramatic benefits.

Later, Mesmer discovered that he could dispense with the magnet; a piece of wood, or even his own hand, would work just as well. From this observation, you or I would conclude that magnetism had nothing to do with the phenomenon. Mesmer, however, drew the quirky conclusion that he did not need a magnet because *he himself* was a magnet. With that claim, he gave us the term “animal magnetism.”

In his later years, Mesmer grew stranger yet. After his death, his followers carried out serious studies of “animal magnetism” or “Mesmerism,” eventually calling it “hypnotism.” But by that time, many physicians and scientists associated hypnosis with eccentrics, charlatans, and other practitioners of hocus-pocus. Even today, hypnosis is part of various stage acts. We must therefore carefully distinguish stage hypnotists from psychologists and psychiatrists who are licensed to practice hypnosis.

Ways of Inducing Hypnosis

Mesmer thought that hypnosis was a power that emanated from his own body, like the power that a magnet exerts on metals. If so, only certain people would have the power to hypnotize others. Today, we believe that becoming a successful hypnotist requires a fair amount of practice but no unusual powers or personality traits.



Although Mesmer is often depicted as irresistibly controlling people, psychologists now recognize that hypnosis reflects a willingness by the hypnotized person, not a special power of the hypnotist.

*Does this sound profound? Or is it nonsense? Mesmer said many things that sound profound at first, but the more we think about them, the less sense they make.



FIGURE 5.14 A hypnotist induces hypnosis by repeating suggestions, relying on the hypnotized person's cooperation and willingness to accept suggestions. No one can force hypnosis on an unwilling person.

There are several ways of inducing hypnosis. The first step toward being hypnotized is simply agreeing to give it a try. Contrary to what you may have seen in the movies or on television, no one can hypnotize an uncooperative person.

A hypnotist might then ask the subject to concentrate, while the hypnotist monotonously repeats such suggestions as, "You are starting to fall asleep. Your eyelids are getting heavy. Your eyelids are getting very heavy. They are starting to close. You are falling into a deep, deep sleep" (Figure 5.14).

In another popular technique, described by R. Udolf (1981), the hypnotist suggests, "After you go under hypnosis, your arm will begin to rise automatically." (Some people, eager for the hypnosis to succeed, shoot their arm up immediately and have to be told, "No, not yet. Just relax; that will happen later.") Then, the hypnotist encourages the subject to relax and suggests that the arm is starting to feel lighter, as if it were tied to a helium balloon. Later, the hypnotist suggests that the arm is beginning to feel a little strange and is beginning to twitch. The timing of this suggestion is important, because when people stand or sit in one position long enough, their limbs really do begin to feel strange and twitch a bit. If the hypnotist's suggestion comes at just the right moment, the subject thinks, "Wow, that's right, my arm does feel a little strange. This is really starting to work!" Wanting to be hypnotized and believing that you are being hypnotized is a big step toward actually being hypnotized.

The Uses and Limitations of Hypnosis

Hypnosis can produce relaxation, concentration, temporary changes in behavior and, sometimes, changes that persist beyond the end of the hypnotic state. A "deeply" hypnotized person will follow many of the hypnotist's suggestions. There is no evidence, however, that hypnosis gives people any new physical or mental abilities.

What Hypnosis Can Do

One well-established effect of hypnosis is to inhibit pain. For some people, a hypnotic suggestion is so effective that they can undergo medical or dental surgery without anesthesia (Figure 5.15). Blocking pain with hypnosis is particularly helpful when people have developed a tolerance for painkilling opiates or if they react unfavorably to anesthetic drugs.

Recall from Chapter 4 that pain has both sensory and emotional components. Hypnosis affects mostly the emotional components. Even when a hypnotized person says that he or she feels no pain, the heart rate and blood pressure still shoot up as much as they do in nonhypnotized people (Hilgard, 1973). Under hypnotic suggestion to feel no unpleasantness, a person subjected to painful stimuli will have high arousal in the parietal cortex areas responsive to body sensations but not in the frontal cortex areas responsive to unpleasant emotions (Rainville, Duncan, Price, Carrier, & Bushnell, 1997). (See Figure 5.16.)

Another constructive use of hypnosis is the **posthypnotic suggestion**, a suggestion that the person will do or experience something particular after coming out of hypnosis. The posthypnotic suggestion could be to do something trivial, such as to scratch one's left ear at exactly 9:00. It could also be something more practical, such as a suggestion to give up tobacco, lose weight, stop nail-biting, break other bad habits, become more sexually responsive, or stop having night terrors (Kihlstrom, 1979; Udolf, 1981). The suggestions do not force any change in behavior; the people who agree to be hypnotized have already decided to try to

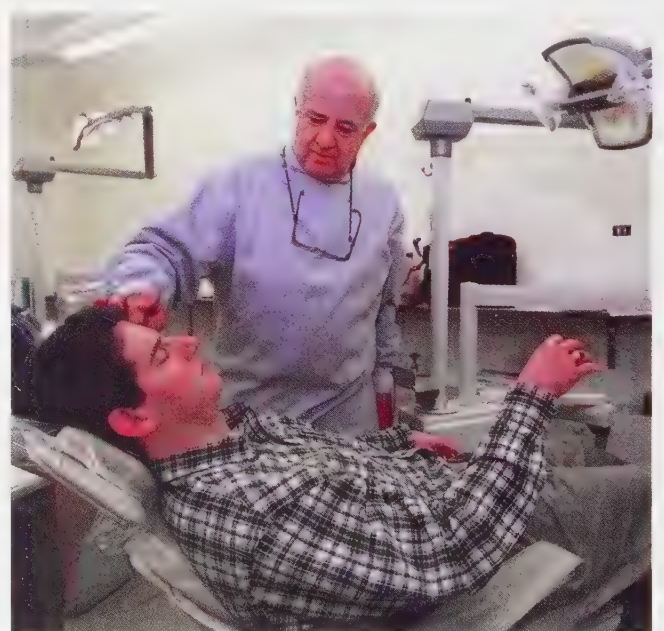


FIGURE 5.15 Some dentists have used hypnosis as their means of relieving pain, even for tooth extractions, root canal surgery, and other seriously painful procedures.

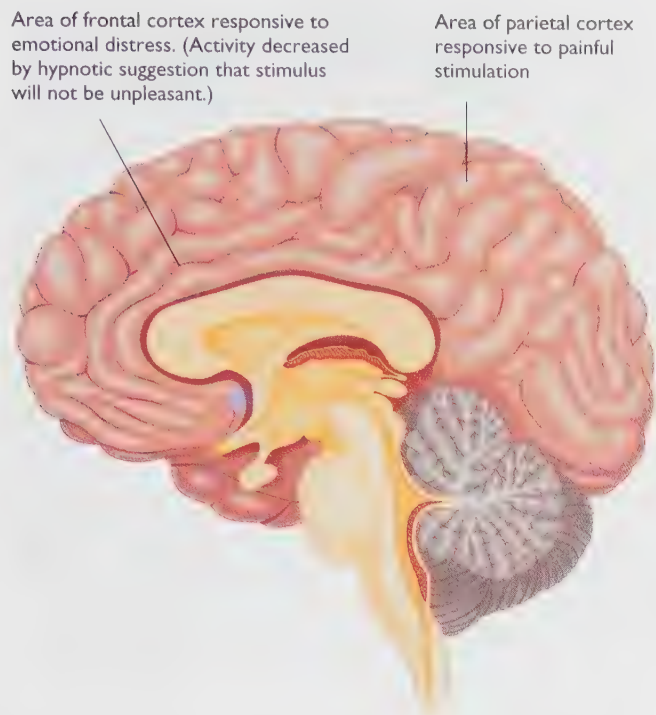


FIGURE 5.16 Hypnotic suggestions to experience less pain can decrease the emotional intensity but generally have less effect on the sensation itself. The hypnotic suggestion that a stimulus will not be unpleasant decreases activity in the frontal cortex areas associated with emotional distress but has little effect on the primary somatosensory cortex.

quit smoking or to change another habit. The hypnosis underscores that resolve.

Posthypnotic suggestions can be effective for helping people to change bad habits, although the effect of a single session wears off in a few days or weeks. In one study, people were taught to use repeated self-hypnosis to quit cigarette smoking. Almost one fourth of those who learned this technique managed to quit smoking for at least the next 2 years (Spiegel, Frischholz, Fleiss, & Spiegel, 1993).

What Hypnosis Cannot Do

Some spectacular claims have been made for the power of hypnotic suggestion, but most of them turn out to be less impressive on closer scrutiny. For instance, people under hypnosis can become as stiff as a board, so stiff that they can balance their head and neck on one chair and their feet on another chair and allow someone to stand on their body (Figure 5.17)! Amazing? Not really. You can probably make

**TRY IT
YOUR-
SELF**

yourself stiff enough to balance this way without being hypnotized. It's easier than it looks. (But I do not recommend that you invite someone to stand on you. A person who does not balance just right could injure you.)

Many people have attempted to use hypnosis to enhance memory. For example, a distressed person tells a psychotherapist, "I don't know why I have such troubles. Maybe I had some really bad experience when I was much younger,

but I just can't remember." Or a witness to a crime says, "I saw the culprit for a second or two, but now my memory is vague and I can't give you a good description." Faced with such situations, therapists and police officers have sometimes turned to hypnotism in the hope of uncovering hidden memories. However, hypnotized people are highly suggestible. Even when they are given an innocent suggestion such as "you will remember more than you told us before," many hypnotized people respond with new but incorrect information. One hypnotized witness confidently reported eating a pizza at a restaurant that does not serve pizza. Another person reported seeing tattoos on a person who had none. A third reported being stabbed repeatedly with a knife but had none of the scars that would result from such an injury (Orne, Whitehouse, Dinges, & Orne, 1988).

In several laboratory studies, experimenters have shown participants either photos or videotapes and then asked questions both without hypnosis and with hypnosis. Overall, hypnosis impaired recall at least as often as it improved it (Steblay & Bothwell, 1994). Furthermore, even long after the end of a hypnotic session, many people expressed great confidence in the inaccurate memories that they "gained" during hypnosis.

In response to such findings, a panel appointed by the



FIGURE 5.17 The U.S. Supreme Court ruled in 1987 that criminal defendants may testify about details they recalled under hypnosis. Its decision sparked this protest by the magician known as "the Amazing Kreskin," who borrowed a stunt usually used to demonstrate the power of hypnosis—standing on a person suspended between two chairs.

American Medical Association (1986) concluded that testimony elicited under hypnosis should not be used in courts of law. The panel voiced no objection to using hypnosis as an investigative tool. For example, if a hypnotized witness reports a license plate number, and the police track down the car and find blood on it, the blood is certainly admissible evidence. However, success stories of that type are rare. Police investigators should therefore disregard a hypnotized witness's report if it does not lead to independent confirmation.

An even more doubtful claim is that hypnosis can help people to recall their early childhood. A hypnotist might say, "You are getting younger. It is now the year _____; now it is _____; now you are only 6 years old." The hypnotized person may give a convincing performance of being a young child, playing with teddy bears and blankets (Nash, Johnson, & Tipton, 1979). But is this person reliving early childhood experiences? Evidently not. *First*, the childhood "memories" that the hypnotized subject so confidently recalls, such as the names of friends and teachers and the details of birthday parties, are generally inaccurate (Nash, 1987). *Second*, a person who has presumably regressed under hypnosis to early childhood retains spelling and other skills learned later in life. When asked to draw a picture, the subject does not draw as children draw but as adults imagine that children draw (Orne, 1951). (See Figure 5.18.) *Third*, hypnotized subjects will respond just as well to suggestions that they are growing older as to suggestions that they are growing younger. They give just as convincing a performance of being an older person and "remembering" events in the future as of being a younger person (Rubenstein & Newman, 1954). Because these people must be acting out an imagined future, we should assume that they are doing the same with the past.

You may even encounter the astounding claim that hypnosis can help someone to recall memories from a previous life. Hypnotized young people who claim to be recollecting a previous life generally describe the life of a person similar to themselves, married to someone who bears an uncanny resemblance to their current boyfriend or girlfriend. If subjects are asked whether their country (in their past life) is at war or what kind of money is in use, their guesses are seldom correct (Spanos, 1987–88).

CONCEPT CHECK

5. List two practical uses of hypnosis that the evidence supports and one that it does not. (Check your answer on page 180.)

The Limits of Hypnosis

Most hypnotists seem to agree about one of the limits of hypnosis: "You don't have to worry," a hypnotist will reassure you. "People never do anything under hypnosis that they

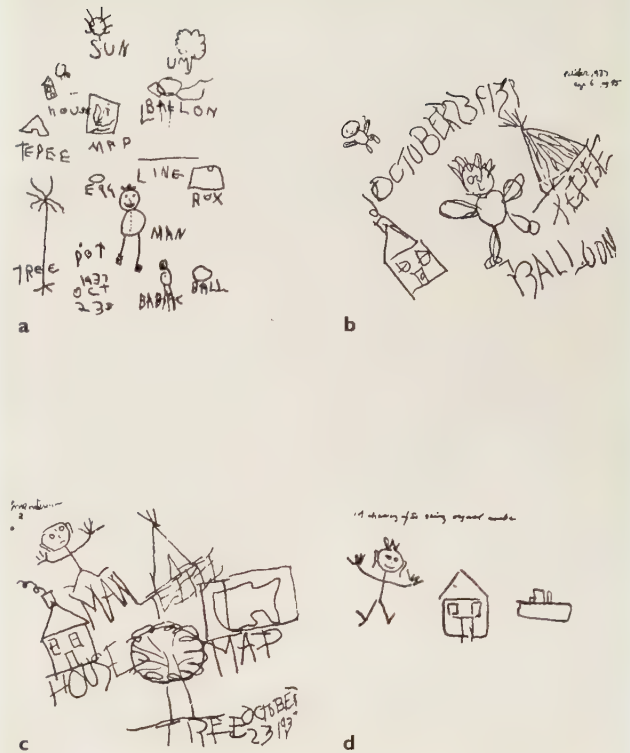


FIGURE 5.18 Regression or role playing? One person made drawing (a) at age 6 and the other three drawings (b, c, & d) as a college student under hypnosis. While under hypnosis, the person was asked to regress to age 6. The drawings created while under hypnosis are not like drawings done in childhood. Orne (1951) concluded that the hypnotized students played the role of a 6-year-old and drew as they thought a child would.

would ordinarily refuse to do." That reassurance is an important strategy in getting you to agree to be hypnotized. But is it true? And how does anyone know whether it is true? Do you suppose that hypnotists have frequently asked clients to perform criminal or immoral acts, so they know that clients consistently refuse? No, of course not. Furthermore, a few investigators have tried to test this statement, but their results are difficult to interpret. The following is one example.

Hypothesis Hypnotized people will sometimes perform acts that nonhypnotized people would refuse to do.

Method Eighteen college students were randomly assigned to three groups. The investigator hypnotized those in one group, instructed those in the second group to pretend they had been hypnotized, and merely asked those in the third group to participate in the study (making no mention of hypnosis). All students were then asked to perform three acts: First, they were told to go to a box in a corner of the room and pick up a poisonous snake. The snake really was poisonous. If a subject got too close to the snake, he or she was restrained at the last moment.



Will hypnotized people do anything that they would otherwise refuse to do? The problem with answering that question is that nonhypnotized people will sometimes perform some strange and dangerous acts, either because an experimenter asked them to or on their own.

Second, the hypnotist poured some highly concentrated, fuming nitric acid into a large container and distinctly stated that it was nitric acid. To dispel any doubts, he threw a coin into the acid and let the subjects watch as it started to dissolve. The hypnotist then told the subjects to reach into the acid with bare hands and remove the coin. Here there was no last-second restraint. Anyone who followed the instructions was told to wash his or her hands in warm soapy water immediately afterward. (Today's ethical procedures would prevent such a study.) Third, the hypnotist instructed the subject to throw the nitric acid into the face of the hypnotist's assistant. Unnoticed, the hypnotist had swapped the container of nitric acid for a container of water, but the hypnotized person had no way of knowing that.

Results Five of the six hypnotized students followed all three directions (Orne & Evans, 1965). Moreover, the six control-group students who were pretending to be hypnotized also followed all three commands! So did two of the six students who were just told to take these actions as part of an experiment, with no mention of hypnosis. (Nonhypnotized subjects did, however, hesitate much longer than the hypnotized subjects.)

Why would people do such extraordinary things? They explained that they simply trusted the experimenter: "If he tells me to do something, it can't really be dangerous."

Interpretation Hypnotized people *will* do some strange things that we assume they would not ordinarily do. However, nonhypnotized people will do the same strange things, at least when they know they are participating in an experiment conducted by someone they regard as reputable.

In short, we simply do not have adequate evidence to decide whether people under hypnosis will do anything

that they would utterly refuse to do otherwise, because nonhypnotized people do things we would have expected them to refuse!

There is a message here about psychological research: If the control groups do not behave as the experimenter expects them to behave, it then becomes very difficult to interpret the behavior of the experimental group.



Distortions of Perception Under Hypnosis

A few people report visual or auditory **hallucinations** (*sensory experiences not corresponding to reality*) under hypnosis; a larger percentage report touch hallucinations (Udolf, 1981). A hypnotist can bring about a touch hallucination by such suggestions as "your nose itches" or "your left hand feels numb."

When hypnotized people say that they see or hear something or that they fail to see or hear something, are they telling the truth or are they just saying what the hypnotist wants them to say? Or do they perhaps believe that they see or hear something, when in fact they do not?

In one experiment to test this question, people who were highly susceptible to hypnosis looked at the Ponzo illusion, shown in Figure 5.19a. Like other people, they reported that the top horizontal line looked longer than the bottom horizontal line. Then they were hypnotized and told not to see the radiating lines but to see only the two horizontal ones. Those who said that they no longer saw the radiating lines still perceived the top line as longer than the

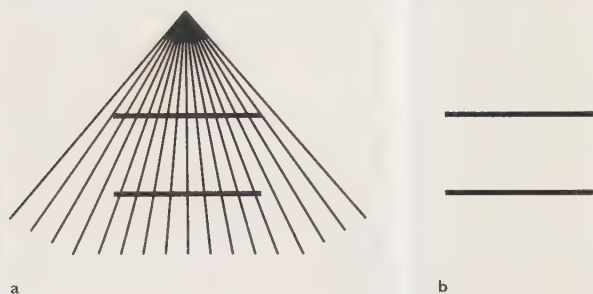


FIGURE 5.19 Horizontal lines of equal length in (a) the Ponzo illusion and (b) without the optical illusion. Researchers employ such visual stimuli to determine how hypnosis may alter sensory perception.

bottom one (Miller, Hennessy, & Leibowitz, 1973). If the radiating lines had truly disappeared, then the subjects would have seen something like Figure 5.19b, where the horizontal lines look equal.

So, does hypnosis alter perception? It does not alter the information that the sensory receptors send to the nervous system. It merely alters how people react to that information and what they do about it.

Is Hypnosis an Altered State of Consciousness?

Hypnosis produces some paradoxical effects. If a hypnotist tells you, “your hand is rising; you can do nothing to stop it,” your hand might indeed rise. If you were later asked why, you might reply (as many do) that you lost control over your own behavior. Still, the behavior is neither reflexive nor involuntary; you raised your hand yourself. Psychologists have wrestled with the issue of how a behavior can be voluntary even though one lacks the usual sense of making a decision (Lynn, Rhue, & Weekes, 1990).

At one extreme, some psychologists regard hypnosis as a special state of consciousness, characterized by greatly increased suggestibility. At the other extreme, other psychologists emphasize the similarities between hypnosis and normal wakeful consciousness, especially that hypnotized people are aware of their surroundings and aware of their own behaviors. Most psychologists take intermediate positions, noting that hypnotized people are neither “faking it” nor under the control of the hypnotist. That is, hypnosis is a special state in some ways and not in others (Kirsch & Lynn, 1998).

One way to determine whether hypnosis is a special state of consciousness is to find out whether un hypnotized people can do everything that hypnotized people do. That is, if you agreed to pretend that you were hypnotized, could you do everything that a truly hypnotized person does?

How Well Can an Unhypnotized Person Pretend to Be Hypnotized?

In several experiments, one group of college students was hypnotized while another group was told to pretend they were hypnotized. An experienced hypnotist then examined all the students and tried to determine which ones were really hypnotized.

Fooling the hypnotist turned out to be easier than expected. The pretenders were able to tolerate sharp pain without flinching and could recall or pretend to recall old memories. They could make their bodies as stiff as a board and lie rigid between two chairs. When standing people were told to sit down, they did so immediately (as hypnotized people do) without first checking to make sure there was a chair behind them (Orne, 1959, 1979). When told to experience anger or another emotion, they exhibited physiological changes such as increased heart rate and sweating, just like hypnotized people do (Damaser, Shor, & Orne, 1963). Not even highly experienced hypnotists could accurately identify the pretenders.

However, a few differences between the hypnotized people and pretenders did emerge (Orne, 1979). The pretenders did certain things differently from the way the hypnotized subjects did them—not because they were unable to do them the same way but because they did not know how a hypnotized subject would act. For instance, when the hypnotist suggested, “you see Professor Schmaltz sitting in that chair,” people in both groups reported seeing the professor. Some of the hypnotized subjects, however, said they were puzzled. “How is it that I see the professor there, but I can also see the entire chair?” Pretenders never reported seeing this “double reality.”

At that point in the experiment, Professor Schmaltz actually walked into the room. “Who is that entering the room?” asked the hypnotist. The pretenders would either say they saw no one, or else they would identify Schmaltz as someone else. The hypnotized subjects would say, “That’s Professor Schmaltz.” Some of them said that they were confused by seeing the same person in two places at the same time. For some of them, the hallucinated professor faded at that moment, whereas others continued to accept the double image.

So . . . what is our conclusion? Is hypnosis an altered state, or are hypnotized people just playing a role? This is not an easy question to answer. Apparently, un hypnotized people playing the role of “hypnotized subjects” can mimic most of the effects of hypnosis, and they probably could have mimicked the other effects if they had known what those effects were. However, the fact that un hypnotized role-players resemble hypnotized people does not necessarily mean that hypnosis is “nothing but” role-playing (Hilgard, 1971). The induction of hypnosis produces a variety of effects that other people must learn to imitate; those effects happen spontaneously for the hypnotized subjects.

Faked Hypnosis by a Criminal Defendant

In 1979 Kenneth Bianchi was arrested for raping and strangling two women. He was suspected of raping and strangling many others in similar fashion; he was suspected of being the “Hillside Strangler” who had been terrifying the Los Angeles area. While he was awaiting trial, a psychiatrist hypnotized him and claimed to uncover a second personality, “Steve Walker,” who had first appeared when Bianchi was 9 years old and who had, Bianchi said, actually committed the crimes. Bianchi pleaded not guilty by reason of insanity.

But was Bianchi really insane, or was he faking the second personality and, indeed, only pretending to be hypnotized (Figure 5.20)? Six psychiatrists were asked to examine Bianchi to try to answer these questions. One of them was Martin Orne, the psychiatrist who had conducted the research on whether college students could effectively pretend to be hypnotized. He knew how difficult it was to detect pretenders, but he had also picked up a few tricks in the course of his earlier research. More importantly, he had cultivated a healthy skepticism. At first, the six psychiatrists were divided about whether Bianchi was really insane or merely faking. Eventually, Orne convinced them that Bianchi was faking, for a variety of reasons including the following (Orne, Dinges, & Orne, 1984):

- Bianchi behaved in ways that were not typical of hypnotized people. For example, when Orne suggested that someone was sitting in an empty chair, Bianchi not only claimed to see that person but even reached out to shake hands! Orne concluded that Bianchi was trying too hard to prove that he was hypnotized.



FIGURE 5.20 Kenneth Bianchi, accused of being the Hillside strangler, pleaded not guilty on the grounds that he had a “multiple personality” and that his evil second personality was responsible for the crimes. Psychiatrist Martin Orne, however, persuaded the court that Bianchi was faking his second personality and, indeed, that he was only pretending to be hypnotized.

- In one hypnosis session, Bianchi’s “Steve” personality tore the filter tip off a cigarette. After the hypnosis was over, the “Ken” personality expressed amazement and said he couldn’t imagine who might have torn the filter tip off. This episode might suggest that Bianchi had completely forgotten his experience under hypnosis. However, Orne observed that Bianchi did exactly the same thing with three other hypnotists. Again, Bianchi was apparently trying to convince everyone that he had been deeply hypnotized.

- At one point, Orne told Bianchi that he doubted Bianchi was a true case of multiple personality because “real” multiple personalities have three personalities, not just two. (This statement is not true; Orne just wanted to see what would happen.) Later that day, Bianchi developed a third personality.

By uncovering these facts, Orne exposed Bianchi’s pretense. Bianchi agreed to plead guilty in return for the state’s dropping its request for the death penalty. He also stopped claiming to have multiple personalities.

In short, it is sometimes possible to distinguish between hypnotized people and those who are only pretending, but it is not easy.

Meditation: In Some Ways Like Hypnosis

Hypnosis is not the only method of inducing a relaxed and possibly altered state of consciousness. **Meditation**, a method of inducing a calm, relaxed state through the use of special techniques, follows a tradition that has been practiced in various parts of the world for thousands of years. For example, one may sit quietly while repeating a simple sound or chant (such as “om”) for 15 to 20 minutes twice a day. The point is “to empty one’s mind”—that is, to expel all the worries, troubles, and doubts that otherwise intrude into our thoughts. In that respect, it has some similarities to the relaxed, passive state of hypnosis. However, meditation is different because it includes no hypnotist or suggestions.

Many studies document moderate benefits from meditation in terms of decreased heart rate, blood pressure, and so forth. These benefits are similar to those produced by other means of achieving prolonged relaxation (Holmes, 1984). As with hypnosis, some practitioners of meditation have made exaggerated claims; however, it is possible to dismiss the exaggerations and still respect the real phenomenon.

THE MESSAGE The Nature of Hypnosis

Researchers have still not reached a consensus on exactly what hypnosis is. They do agree on a few general points, however: Hypnosis is not just faking or pretending to be



Meditation excludes the worries and concerns of the day and thereby induces a calm, relaxed state.

hypnotized, and it does not give people mental or physical powers that they otherwise lack. Hypnosis merely enables people to relax, concentrate, and follow suggestions better than they usually do. Be skeptical of anyone who claims much more than that for hypnosis.

SUMMARY

- ★ **Nature of hypnosis.** Hypnosis is a condition of increased suggestibility that occurs in the context of a special hypnotist-subject relationship. Psychologists try to distinguish the genuine phenomenon, which deserves serious study, from exaggerated claims. (page 173)
- ★ **Hypnosis induction.** To induce hypnosis, a hypnotist asks a person to concentrate and then makes repetitive suggestions. The first steps toward being hypnotized are to be willing to be hypnotized and to believe that one is becoming hypnotized. (page 173)
- ★ **Uses.** Hypnosis can alleviate pain, and through posthypnotic suggestion, it can help someone overcome bad habits, at least temporarily. (page 174)
- ★ **Nonuses.** Hypnosis does not give people special strength or unusual powers. Most of the new “memories” evoked under hypnosis are false. (page 175)
- ★ **Uncertain limits.** Although many hypnotists insist that hypnotized people will not do anything that they would refuse to do when not hypnotized, there is little solid evidence to back up this claim. (page 176)

- ★ **Sensory distortions.** People under hypnosis can be induced to ignore certain stimuli as if they were blind or deaf. However, the ignored information still influences behavior in subtle or indirect ways. (page 177)
- ★ **Hypnosis as an altered state.** Controversy continues about whether hypnosis is a special state of consciousness and, indeed, what we mean by an altered state. (page 178)
- ★ **Meditation.** Meditation can induce a relaxed condition that one might regard as an altered state of consciousness. (page 179)

Suggestion for Further Reading

Rhue, J. W., Lynn, S. J., & Kirsch, I. (Eds.). (1993). *Handbook of clinical hypnosis*. Washington, DC: American Psychological Association. A collection of articles, mostly about the uses of hypnosis for treating psychological disorders.

Terms

- hypnosis** a condition of increased suggestibility that occurs in the context of a special hypnotist-subject relationship (page 173)
- posthypnotic suggestion** a suggestion made to hypnotized subjects that they will do or experience something particular after coming out of hypnosis (page 174)
- hallucination** a sensory experience not corresponding to reality, such as seeing or hearing something that is not present or failing to see or hear something that is present (page 177)
- meditation** a method of inducing a calm, relaxed state through the use of special techniques (page 179)

Answer to Concept Check

5. Supported by the evidence: Hypnosis can decrease pain, and posthypnotic suggestions can help people break bad habits. Not supported: Hypnosis does not improve people's memories. (Neither does it give them added strength.) (page 176)

Web Resources

Hypnosis

www.psych-web.com/asc/hyp.html

A collection of FAQs includes Todd Stark on the nature of the hypnotic trance, accuracy of memory, hypnotizability, enhanced mental and physical functioning, and other topics; also articles on “Social Reconstruction of Memories,” “Hypnosis in Pain Management,” and “Past-Life Hypnotic Regression.”

Drugs and Their Effects

What experiences do abused drugs produce?

Why do people experiment with such drugs?

Many people assume (incorrectly) that any drug they receive from a physician or from a pharmacy must be safe. In fact, almost any drug is dangerous in large doses, and many legal drugs can become habit-forming after repeated use. Furthermore, many abused drugs—including amphetamines, morphine, and even cocaine—have legitimate medical uses. The dividing line between “good drugs” and “bad drugs” is a blurry one; it depends more on the quantities used and the reasons for their use than it does on the chemistry of the drugs themselves.

The abuse of alcohol and other drugs is a widespread problem, which we shall consider in Chapter 16. Here, our focus is on the experiences associated with common drugs of abuse. I have tried to present only information that is backed by solid evidence, but in the study of drug effects, it is more difficult than usual to separate fact from fiction. Many people who are trying to discourage drug use have made strong claims about the harms that result from using various drugs. For example, although some drugs—such as MDMA (“ecstasy”) and PCP (“angel dust”)—can apparently produce brain damage, there is little evidence to support claims of permanent damage from using other drugs (McCann, Lowe, & Ricaurte, 1997). The risk of presenting exaggerated claims is that people stop believing the warnings. At the opposite extreme, many people who defend their own use of drugs have minimized the dangers; again, many of their claims go beyond the facts. As with other topics, we should be skeptical of claims that go beyond the evidence.

A Survey of Abused Drugs and Their Effects

Some abused drugs, such as alcohol and opiates, have predominantly calming effects. Others, such as amphetamines and cocaine, are known for their stimulating effects. Still others, such as LSD, produce hallucinations. Table 5.2 lists some of the commonly abused drugs and their

TABLE 5.2 Commonly Abused Drugs and Their Effects

DRUG CATEGORY	EFFECTS ON BEHAVIOR	EFFECTS ON CENTRAL NERVOUS SYSTEM AND ORGANS
Depressants Alcohol	Relaxant; relieve inhibitions; impair memory and judgment	Widespread effects on membranes of neurons; facilitate activity at GABA synapses
Tranquilizers Barbiturates; benzodiazepines (Valium, Xanax)	Relieve anxiety; relax muscles; induce sleep	Facilitate activity at GABA synapses
Opiates Morphine, heroin	Decrease pain; decrease attention to real world; unpleasant withdrawal effects as drug leaves synapses	Stimulate endorphin synapses
Stimulants Caffeine	Increase energy, alertness	Increase heart rate; indirectly increase activity at glutamate synapses
Amphetamines Cocaine	Increase energy, alertness	Increase or prolong activity at dopamine synapses
Mixed Stimulant-Depressants Nicotine	Stimulate brain activity, but most smokers say cigarettes relax them	Stimulate activity at some (not all) acetylcholine synapses; increase heart rate
Distortion of Experience Marijuana (THC)	Intensifies sensory experiences; distorts perception of time; can relieve glaucoma, nausea; sometimes impairs learning, memory	Attaches to receptors found in hippocampus and other areas; normal role of these receptors is unknown
Hallucinogens LSD; mescaline	Cause hallucinations, sensory distortions, and occasionally panic	Alter pattern of release and binding of serotonin

most prominent effects. Do they have anything in common that would account for their tendency to be abused? Apparently, yes. Nearly all abused drugs increase the activity at dopamine synapses in the brain, especially those with dopamine receptor types D_2 , D_3 , and D_4 at the postsynaptic membrane (Maldonado et al., 1997). Some drugs (such as amphetamines) increase dopamine release directly; others do so indirectly by stimulating certain synapses which in turn stimulate dopamine release. Of the few abused drugs that are known not to increase dopamine activity, such as phencyclidine (PCP, or “angel dust”), the general pattern is that they inhibit activity at glutamate receptors (Carlezon & Wise, 1996). In a small brain area known as the *nucleus accumbens*, which is apparently critical for either the motivating or the attention-getting effects of experience, glutamate is an excitatory transmitter and dopamine is inhibitory. Thus, drugs that inhibit glutamate (such as PCP) and drugs that stimulate dopamine (most other abused drugs) have the same net effect on the nucleus accumbens. Figure 5.21 shows the location of the nucleus accumbens.

Alcohol

When archeologists unearthed a Neolithic village in Iran's Zagros Mountains, they found a jar that had been constructed about 5400–5500 B.C., which was one of the oldest human-made crafts ever found (Figure 5.22). Inside the



FIGURE 5.22 This jar, dated about 5400–5500 B.C., is one of the oldest human crafts ever found. It was used for storing wine.

jar, especially near the bottom, the archèologists found a yellowish residue. They were curious to know what the jar had held, so they sent some of the residue for chemical analysis. The unambiguous answer came back: The jar had served as a vessel for wine (McGovern, Glusker, Exner, & Voigt, 1996).

Clearly, human use of alcohol is a tradition that has stood the test of time. **Alcohol** is a *class of molecules that includes methanol, ethanol, propyl alcohol (rubbing alcohol), and others*. Ethanol is the type that people drink; the others are highly dangerous if consumed. Alcohol acts primarily as a relaxant. It can lead to heightened aggressive, sexual, and risk-taking behaviors, mainly by depressing the brain areas that ordinarily inhibit such behaviors. Moderate use of alcohol serves as a tension reducer and a social lubricant. It helps people to forget their problems, at least for the moment.

Excessive use can damage the liver and other organs, aggravate or prolong many medical conditions, and impair memory and motor control. A woman who drinks alcohol during pregnancy risks impairment to her baby's brain development (see Chapter 10).

Alcohol abuse occurs throughout the world, although it is more common in some populations than others. Within the United States, alcohol abuse is more common among Native Americans than among other ethnic groups, more widespread among people of African ancestry than those of European ancestry, and more common among people of European ancestry than those of Asian ancestry. The explanation for these ethnic differences is unknown, but it is not true (as many have supposed) that Native Americans get drunk more easily than others. If anything, they are *less* affected than others by a moderate amount of alcohol and therefore continue drinking (Garcia-Andrade, Wall, & Ehlers, 1997). Whatever the explanation, these ethnic differences are worth taking seriously; many observers believe that differences in alcohol use constitute a major reason for

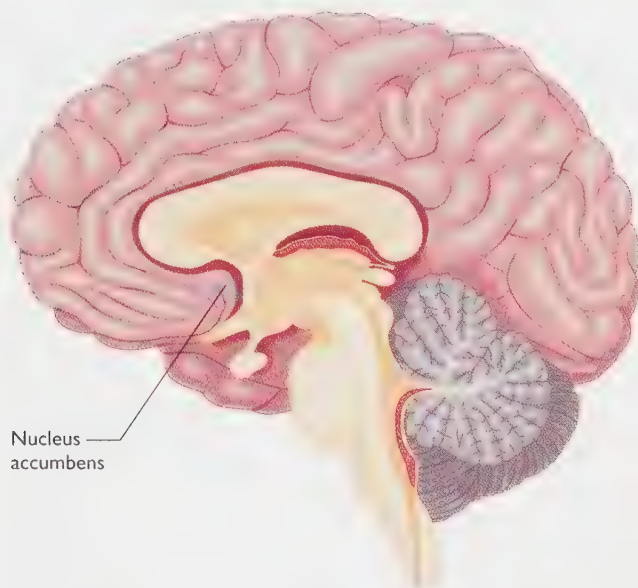


FIGURE 5.21 The nucleus accumbens is a small brain area that is critical for the motivating effects of many experiences, including drugs, food, and sex. Most abused drugs increase the activity of dopamine, an inhibitory transmitter in this area. Other abused drugs, such as PCP, inhibit the activity of glutamate, an excitatory transmitter here. That is, a decrease in output by the nucleus accumbens is important for the effects of abused drugs and other motivating experiences.

the shorter average life spans of Native Americans and African-Americans, as compared with other groups (Rivers, 1994).

Tranquilizers

Tranquilizers help people to relax and to fall asleep; they can also decrease muscle tension and suppress epileptic seizures. Barbiturates, once a common type of tranquilizer, proved to be highly habit-forming and easily fatal in large doses. Today, the most commonly used tranquilizers are a class of chemicals called *benzodiazepines*, which include the drugs diazepam (Valium) and alprazolam (Xanax). Benzodiazepines can also be habit-forming, although less so than barbiturates. Thousands of tons of these drugs are taken in pill form every year in the United States.

Benzodiazepines exert their calming effects by facilitating transmission at synapses that use the neurotransmitter GABA. Alcohol facilitates transmission at the same synapses, though by a different mechanism (Sudzak et al., 1986). Taking alcohol and tranquilizers together can be dangerous, because together they increase GABA transmission more effectively than either alcohol or tranquilizers could alone and thereby suppress activity in the areas of the medulla that control breathing and the heartbeat.

One benzodiazepine drug, flunitrazepam, has attracted attention as the “date rape drug.” Flunitrazepam dissolves quickly in water; it has no color, odor, or taste; and it can therefore be easily slipped into a woman’s drink without her realizing it. The effects of this drug are similar to those of other tranquilizers; these effects include drowsiness, poor muscle coordination, and memory impairment, which can impair a woman’s later ability to describe the attack (Anglin, Spears, & Hutson, 1997; Woods & Winger, 1997). Although flunitrazepam apparently has been used to facilitate date rape, there is no evidence that this drug is fundamentally different from, or more dangerous than, other benzodiazepine tranquilizers, however. When a hospital suspects that a woman has been given flunitrazepam, the hospital can run a simple urine test to determine the presence of this drug.

Opiates

Opiates are either natural drugs derived from the opium poppy or synthetic drugs with a chemical structure similar to that of natural opiates. An opiate drug makes people feel happy, warm, and content, with little anxiety or pain. However, the person will experience nausea and will tend to ignore the real world. Once the drug has left the brain, the affected synapses become understimulated, and the user then enters withdrawal. Elation gives way to anxiety, pain, and hyperresponsiveness to sounds and other stimuli.

Morphine (named after Morpheus, the Greek god of dreams) has important medical use as a painkiller. When used for that purpose, morphine is almost never habit-forming.

Here is an important point: No drug is automatically habit-forming. The probability of abuse or habit formation depends partly on the drug but largely on the person taking it and their reasons for doing so.

Opiate drugs such as morphine, heroin, methadone, and codeine bind to a specific set of neurotransmitter receptors in the brain (Pert & Snyder, 1973). The discovery of neurotransmitter receptors prompted neuroscientists to look for naturally occurring brain chemicals that bind to those receptors, because it hardly seemed likely that evolution would equip us with receptors to respond just to extracts of the opium poppy. Researchers found that the brain produces several chemicals, called **endorphins**, that *bind to the opiate receptors* (Hughes et al., 1975). Endorphins serve to inhibit chronic pain.

Marijuana

Marijuana produces a variety of effects, which can include drowsiness, an intensification of sensory experiences, and the illusion that time is passing very slowly (Weil, Zinberg, & Nelson, 1968). It can have certain medical uses, although its medical applications are still controversial. For example, it is sometimes used to reduce pressure in the eyes in an attempt to relieve glaucoma, a common cause of blindness. It also reduces nausea, acts as a weak painkiller, and suppresses tremors, although it is uncertain whether marijuana is ever a better treatment than other medications that have less potential for abuse.

Although people are aware of marijuana’s effects for no more than 2 or 3 hours after using it, more subtle effects can



After California legalized marijuana for medical uses, many clubs and stores opened for the sale and distribution of the drug.

persist much longer. Marijuana dissolves in the fats of the body, so traces of it can be found for weeks after the drug has been used (Dackis, Pottash, Annitto, & Gold, 1982).

The literature on marijuana seems to be dominated by people exaggerating the harm of the drug, to discourage people from using it, and people claiming that the drug has almost no dangers, to argue for the reduction or elimination of criminal penalties for using it. Sometimes it is difficult to discern the truth.

One of the arguments emphasizing the dangers of marijuana is that many users are criminals, mentally ill, or otherwise disreputable people. However, correlation does not demonstrate causation. We have no more reason to say that marijuana leads to criminal behavior or mental illness than to say that people with a predisposition toward criminality or mental illness are likely to use marijuana. Many have also attacked marijuana as a “gateway drug,” on the grounds that many people who eventually use heroin or cocaine used marijuana first. True, but again we cannot infer cause and effect. Before using heroin, cocaine, or marijuana, most had also used alcohol and tobacco; many had experimented with risky sexual behaviors; and so on. Some of the those who try marijuana, alcohol, or tobacco go on to try heroin or cocaine, but others do not.

On the other hand, marijuana use certainly does pose some risks. Frequent smoking of marijuana increases the risk of lung cancer, just as tobacco smoking does. Many users experience impairment of learning and memory (Miller & Branconnier, 1983). Animal research indicates that marijuana smoke can temporarily shrink the dendrites of brain neurons (Westlake et al., 1991). In short, marijuana use is not enormously risky, but it is hardly risk-free, either.

The active ingredient in marijuana (*cannabis*) is THC, or tetrahydrocannabinol, which attaches to receptors that are abundant throughout the brain (Herkenham, Lynn, deCosta, & Richfield, 1991), especially in the hippocampus (an important brain area for memory) and brain areas important for the control of movement (Herkenham et al., 1990). The presence of those receptors implies that the brain produces some THC-like chemical of its own. Researchers first discovered a brain chemical that they named *anandamide* (from *ananda*, the Sanskrit word for “bliss”) that attaches to the same receptors as THC (Devane et al., 1992). However, the brain has very low levels of anandamide, compared to the abundant THC receptors. A more prevalent compound, discovered later, that attaches to the same receptors, has the not-very-catchy name *sn-2 arachidonylglycerol*, abbreviated 2-AG (Stella, Schweitzer, & Piomelli, 1997). At this point, no one knows the functions of anandamide or 2-AG. In short, we do not yet know why your brain manufactures its own marijuana-like chemicals. Curiously, chocolate also contains three chemicals that probably also attach to the THC receptors (diTomaso, Beltramo, & Piomelli, 1996). So, people who say that they are “addicted to chocolate” may be more literally correct than they had supposed.

The research on THC receptors helps to explain why people frequently die of an overdose of opiates, but rarely of marijuana: Opiate receptors are densely located in the medulla and other brain areas that control heart rate and breathing, whereas these same areas have very few THC receptors (Herkenham et al., 1990). So even rather large doses of marijuana are unlikely to stop the heart or to interfere with breathing.

Stimulant Drugs

Stimulants are drugs that boost energy, heighten alertness, increase activity, and produce a pleasant feeling. Coffee, tea, and many soft drinks contain caffeine, which is a stimulant. People who drink a lot of coffee become dependent on caffeine; if someone replaces their regular coffee with decaffeinated coffee, they will experience headaches and drowsiness (Hughes et al., 1991). Surgical patients sometimes report various kinds of distress that can be mistaken for the effects of their operation, when in fact these patients are suffering from caffeine withdrawal.

Amphetamines and cocaine are powerful stimulants with wide-ranging effects. Both drugs prevent neurons from reabsorbing the dopamine they have released; they thereby prolong the effects of the dopamine (Ritz, Lamb, Goldberg, & Kuhar, 1987). Because both of these drugs also increase the activity at norepinephrine and serotonin synapses, their effects on behavior are complex. Both drugs increase the heart rate, blood pressure, and body temperature. Cocaine has additional anesthetic (sensation-blocking) effects, similar to the effects of novocaine and lidocaine.

We regard cocaine as a stimulant because it increases heart rate, makes people excited and alert, and interferes with their sleep. However, cocaine actually *decreases* the overall activity within the brain (London et al., 1990). That effect may seem to contradict the statement that cocaine prolongs the activity of a couple of neurotransmitters; however, those transmitters are predominantly inhibitory transmitters. Thus, by increasing dopamine and norepinephrine activity, cocaine decreases the activity of many brain neurons (Figure 5.23).

If cocaine decreases the activity of neurons, you might ask, how does it act as a stimulant for behavior? The brain is a complicated organ that often operates on the principle of double negatives: Cocaine decreases the activity of neurons, which were in turn acting to inhibit still other neurons. By inhibiting an inhibitor, cocaine has the net effect of stimulating the final behavioral outcome.

Cocaine has long been available in the powdery form of cocaine hydrochloride, a chemical that can be sniffed. It produces mostly enjoyable effects that increase gradually over a few minutes and then decline gradually over about half an hour. It also anesthetizes the nostrils and can in some cases damage the lungs. Sniffed cocaine hydrochloride is only occasionally habit-forming.

Before 1985, the only way to get a more intense effect

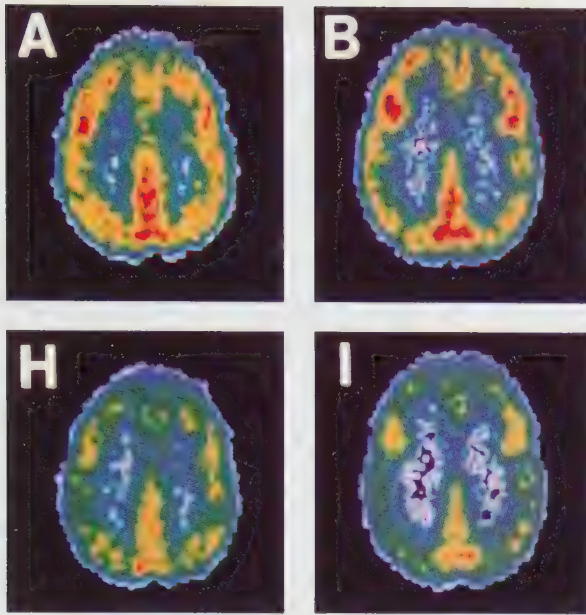


FIGURE 5.23 “Your brain on drugs.” Parts A and B show the activity of a normal brain in horizontal section, as measured by PET scans. Parts H and I show activity of the same brain under the influence of cocaine. Red indicates the highest amount of activity, followed by yellow, green, and blue. Note that cocaine has decreased the amount of activity in the brain.

from cocaine hydrochloride was to treat it with ether to convert it into *free-base cocaine*—cocaine with the hydrochloride removed. Smoking free-base cocaine enables a high percentage of it to enter the body rapidly and thereby enter the brain rapidly. The faster a drug enters the brain, the more intense the resulting experience will be.

The drug known as *crack cocaine* first became available in 1985. Crack is cocaine that has already been converted into free-base rocks, ready to be smoked (Brower & Anglin, 1987; Kozel & Adams, 1986). (See Figure 5.24.) It is called “crack” because it makes popping noises when smoked. Crack produces a rush of potent effects within just a few seconds, much faster than other forms of cocaine. The effects are generally described as pleasant, although some people report intense anxiety instead, and other people suffer heart attacks or other severe medical complications. Long-term use can lead to a sore throat, mental confusion, lung diseases, and other serious problems.

Because crack cocaine enters the brain so rapidly, it can become powerfully habit-forming, although the habit forms gradually over two to four years, so frequent users may lull themselves into a false sense of security: “I can take it or leave it” (Gawin, 1991). During periods of using the drug, the drug experience itself becomes the focus of so much attention that the person neglects usual activities such as eating, sleeping, going to work, or taking care of family members.

Because selling crack is so lucrative, rival gangs in large cities compete with each other to control the sales. The resulting violence has created a problem for society that goes far beyond the direct harm done by the drug itself.



FIGURE 5.24 Because crack is smoked, it reaches the brain in 8 seconds, much faster than other forms of cocaine. All else being equal, the faster that a drug reaches the brain, the more intense the experience will be and the greater the probability of addiction.

Tobacco cigarettes deliver nicotine, a chemical that is generally classed as a stimulant although it has complex effects on the body (Stolerman, 1991). Although nicotine increases the heart rate and blood pressure, with many people it also decreases breathing rate (Jones, 1987). Perhaps because of this decreased breathing rate, many smokers say that they find cigarettes relaxing. Nicotine attaches to receptors that increase the release of dopamine in the nucleus accumbens; thus, its effects overlap those of amphetamines, cocaine, and other addictive drugs (Levin & Rose, 1995; Pontieri, Tanda, Orzi, & DiChiara, 1996). Nicotine also stimulates acetylcholine receptors. Nicotine addiction develops relatively slowly, compared to cocaine or opiate addiction, but once it develops, it is very persistent and very hard to break. We shall consider cigarette addiction in more detail in Chapter 16.

Hallucinogens

Drugs that induce sensory distortions are called **hallucinogens** (Jacobs, 1987). Most of these drugs are derived from certain mushrooms or other plants (Figure 5.25); some are manufactured in chemistry laboratories. The hallucinogenic drugs LSD, PCP, and mescaline intensify sensations and can sometimes produce a dreamlike state or an